

Submission Requirements

Required Submission Bundle

Each group must submit the following **five** components:

1. PostgreSQL schema definition (`.sql` file)
2. ETL notebook (`.ipynb` file)
3. LLM prompt document (`.md` file)
4. Supporting PDF
5. Group peer-assessment file

1. PostgreSQL Schema Definition (`.sql` file)

A standalone SQL file that creates all tables, constraints, indexes, and other objects required for your relational model.

Requirement: The evaluator must be able to reproduce your PostgreSQL schema from the submitted `.sql` file without modification.

2. ETL Notebook (`.ipynb` file)

A Jupyter notebook that performs the full extraction, transformation, and loading process from your MongoDB collections into PostgreSQL.

- The **first cell** must clearly list all required Python packages.
- Preferably include a `requirements.txt` section that reflects the exact environment used to run the notebook.

Requirement: The evaluator must be able to run your ETL notebook end-to-end, loading data from MongoDB into PostgreSQL to answer your analytical questions.

3. LLM Prompt Document (`.md` file)

A short document describing your data model in a way that an LLM can understand.

The prompt must not exceed 1,000 words, excluding any code examples. You are strongly encouraged to test and refine it using AI tools.

The prompt will be evaluated using Claude Sonnet 4.6 with the following template:

```
"You are an expert in data analytics. The description of the data model is: {{your prompt document}}. I would like you to answer this analytical question: {{my chosen question}}. Please provide SQL code only. The code should clearly output the answer along with any additional columns you consider necessary."
```

For better performance your prompt should explicitly state:

- the grain of each table
 - the main join keys
 - caveats that affect counts or grouping (e.g. packages with no versions, advisories with no affected ranges)
 - one or two short join examples where they help clarify the model
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4. Supporting PDF

A concise, well-structured PDF containing:

- Your data-model design documentation
- The analytical questions your group selected (one per group member)
- The SQL queries written for those questions, together with the results they produced
- A brief contribution summary for each group member

The single PDF must be no more than 2,000 words, not counting SQL code. LLM use is allowed and encouraged, but the final submission must clearly demonstrate human-in-the-loop involvement.

Data-model design documentation must clearly describe:

- The PostgreSQL design and why it fits the workload
 - How the MongoDB raw collections were ETL'd into PostgreSQL
 - Entity grain for each table
 - All primary keys, foreign keys, or logical join keys
 - Limitations, exclusions, and known edge cases
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5. Group Peer-Assessment File

A short document summarising each member's contribution and peer evaluation.